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Conditioning & Transformations of a RV

EEE 554 — Lecture #6

Conditioning a RV on an Event

Let A be an event defined in terms of a random variable (RV) x ; e.g., $A = \{0 < x < 1\}$ or $A = \{|x| < 5\}$. The conditional PDF of x given A is:

$$f_x(x|A) = \begin{cases} \frac{f_x(x)}{P(A)} & x \in A \\ 0 & x \notin A \end{cases}$$

Note that $f_x(x|A) \geq 0$ for all x and

$$\int_{-\infty}^{\infty} f_x(x|A)dx = \frac{1}{P(A)} \int_A f_x(x)dx = 1,$$

so $f_x(x|A)$ is a valid PDF.

Conditional Moments

Moments are defined in the usual way with respect to the conditional PDF:

$$E[x|A] = \int_{-\infty}^{\infty} x f_x(x|A)dx$$

And the expectation rule applies:

$$E[g(x)|A] = \int_{-\infty}^{\infty} g(x) f_x(x|A)dx$$

So,

$$E[x^n|A] = \int_{-\infty}^{\infty} x^n f_x(x|A)dx$$

and

$$\text{var}(x|A) = E[x^2|A] - (E[x|A])^2$$

Example:

$$f_x(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{else} \end{cases}$$

Let $A = \{x > \frac{1}{2}\} \implies P(A) = \frac{3}{4}$.

The conditional PDF is:

$$f_x(x|A) = \begin{cases} \frac{8}{3}x & \frac{1}{2} < x \leq 1 \\ 0 & \text{else} \end{cases}$$

Calculating conditional moments:

$$E[x|A] = \frac{8}{3} \int_{1/2}^1 x^2 dx = \frac{8}{9} x^3 \Big|_{1/2}^1 = \frac{8}{9} \left(1 - \frac{1}{8}\right) = \frac{7}{9}$$

$$E[x^2|A] = \frac{8}{3} \int_{1/2}^1 x^3 dx = \frac{2}{3} x^4 \Big|_{1/2}^1 = \frac{2}{3} \left(1 - \frac{1}{16}\right) = \frac{5}{8}$$

$$\text{var}(x|A) = \frac{5}{8} - \left(\frac{7}{9}\right)^2 = \frac{5}{8} - \frac{49}{81} = \frac{13}{648}$$

Transformation of One RV

Suppose $g(x)$ is a monotonically increasing function. Define a new RV $y = g(x)$. Then:

$$\begin{aligned} F_y(y) &= P(y \leq y) = P(g(x) \leq y) \\ &= P(x \leq g^{-1}(y)) = F_x(g^{-1}(y)) \end{aligned}$$

where g^{-1} is defined so that $g(g^{-1}(y)) = y$.

Note that by the chain rule:

$$\frac{d}{dy} g(g^{-1}(y)) = g'(g^{-1}(y)) \cdot (g^{-1})'(y) = 1$$

So,

$$(g^{-1})'(y) = \frac{1}{g'(g^{-1}(y))}$$

Thus, the PDF of y is:

$$\begin{aligned} f_y(y) &= \frac{d}{dy} F_y(y) = \frac{d}{dy} F_x(g^{-1}(y)) \\ &= f_x(g^{-1}(y)) \cdot (g^{-1})'(y) \\ &= \frac{f_x(x)}{g'(x)} \Big|_{x=g^{-1}(y)} \end{aligned}$$

Example: $x \sim U[-1, 1]$ (Uniform), and $y = g(x) = x^3$. We have $g^{-1}(y) = y^{1/3}$ and $g'(x) = 3x^2$.

$$f_y(y) = \frac{(1/2)}{3x^2} \Big|_{x=y^{1/3}} = \frac{1}{6} y^{-2/3}$$

for $-1 < y < 1$ and zero otherwise.

Note that it integrates to 1:

$$\int_{-\infty}^{\infty} f_y(y) dy = \frac{1}{6} \int_{-1}^1 y^{-2/3} dy = \frac{1}{2} y^{1/3} \Big|_{-1}^1 = 1$$

With this PDF for y , direct calculation yields:

$$E[y^2] = \int_{-\infty}^{\infty} y^2 f_y(y) dy = \frac{1}{6} \int_{-1}^1 y^{4/3} dy = \frac{1}{7}$$

Using the expectation rule with f_x gives the same result:

$$E[y^2] = E[x^6] = \frac{1}{2} \int_{-1}^1 x^6 dx = \frac{1}{7}$$

If g is monotonically **decreasing**, a similar calculation to the one above gives:

$$f_y(y) = \frac{f_x(x)}{|g'(x)|} \Bigg|_{x=g^{-1}(y)}$$

If g is not strictly monotonic everywhere but is monotonic on a sequence of disjoint intervals $A_k = [a_k, b_k)$, then by total probability:

$$f_y(y) = \sum_k f_y(y|A_k)P(A_k)$$

where each $f_x(x|A_k)$ is monotonic. The expression may be applied to each of these conditional PDFs individually, which can be combined to obtain $f_y(y)$.

Linear Transformation of a Normal RV

Let $x \sim N(\mu, \sigma^2)$ and set:

$$y = g(x) = ax + b$$

with $a \neq 0$. Then $g^{-1}(y) = \frac{y-b}{a}$ and $|g'(x)| = |a|$.

Using the transformation rule:

$$\begin{aligned} f_y(y) &= \frac{f_x(x)}{|a|} \Bigg|_{x=\frac{y-b}{a}} \\ f_y(y) &= \frac{1}{|a|} \cdot \frac{1}{\sqrt{2\pi}\sigma^2} \exp \left\{ -\frac{1}{2\sigma^2} \left(\frac{y-b}{a} - \mu \right)^2 \right\} \\ &= \frac{1}{\sqrt{2\pi}(a^2\sigma^2)} \exp \left\{ -\frac{1}{2(a^2\sigma^2)} (y - (a\mu + b))^2 \right\} \end{aligned}$$

So, $y \sim N(a\mu + b, a^2\sigma^2)$; i.e., a linear transformation on a normal RV produces another normal RV.

Even without calculating f_y explicitly, it is obvious that:

$$E[y] = a\mu + b \quad \text{and} \quad \text{var}(y) = a^2\sigma^2$$

Hence, knowing y is normal immediately gives its PDF.